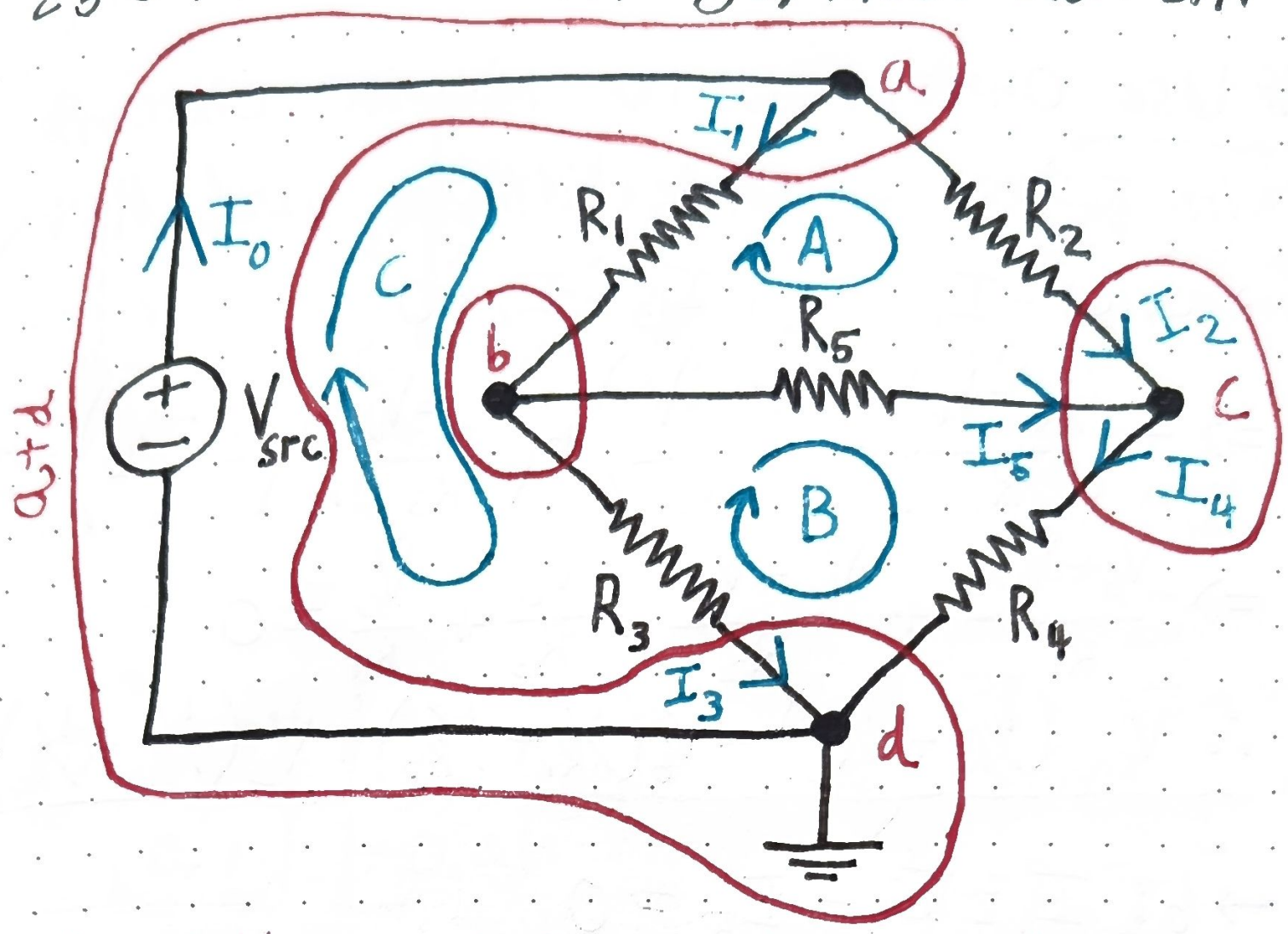
250917 Node-Voltage/Mesh-current



Node Voltage: Extraordinary Nodes – a, b, c, & d

→ node d set as ground

→ we know $V_a = V_{src}$ and $V_d = 0\,V$

→ unknowns are V_b and V_c

* Apply KCL at nodes (and super-nodes):

→ Can't use ohm's law for I_0, thus combine a and d making a+d

→ $a+d: -I_1 - I_2 + I_3 + I_4 = 0$

→ $b: I_1 - I_3 - I_5 = 0$

→ $c: I_2 - I_4 + I_5 = 0$

* Use Ohm's law to rewrite currents in terms of node voltages and R's:

→ a+d: $-I_1 - I_2 + I_3 + I_4 = 0$

$$\Rightarrow -\left(\frac{\cancelto{V_{src}}{V_a} - V_b}{R_1}\right) - \left(\frac{\cancelto{V_{src}}{V_a} - V_c}{R_2}\right) + \left(\frac{V_b - \cancelto{0}{V_d}}{R_3}\right) + \left(\frac{V_c - \cancelto{0}{V_d}}{R_4}\right) = 0$$

$$\Rightarrow \frac{V_b - V_{src}}{R_1} + \frac{V_c - V_{src}}{R_2} + \frac{V_b}{R_3} + \frac{V_c}{R_4} = 0$$

$$\therefore V_{src}\left(1/R_1 + 1/R_2\right) = V_b\left(1/R_1 + 1/R_3\right) + V_c\left(1/R_2 + 1/R_4\right)$$

→ b: $I_1 - I_3 - I_5 = 0$

$$\Rightarrow \left(\frac{\cancelto{V_{src}}{V_a} - V_b}{R_1}\right) - \left(\frac{V_b - \cancelto{0}{V_d}}{R_3}\right) - \left(\frac{V_b - V_c}{R_5}\right) = 0$$

$$\Rightarrow \frac{V_{src} - V_b}{R_1} + \frac{-V_b}{R_3} + \frac{V_c - V_b}{R_5} = 0$$

$$\therefore V_{src}/R_1 = V_b\left(1/R_1 + 1/R_3 + 1/R_5\right) + V_c\left(-1/R_5\right)$$

→ c: $I_2 - I_4 + I_5 = 0$

$$\Rightarrow \left(\frac{\cancelto{V_{src}}{V_a} - V_c}{R_2}\right) - \left(\frac{V_c - \cancelto{0}{V_d}}{R_4}\right) + \left(\frac{V_b - V_c}{R_5}\right) = 0$$

$$\Rightarrow \frac{V_{src} - V_c}{R_2} + \frac{-V_c}{R_4} + \frac{V_b - V_c}{R_5} = 0$$

$$\therefore V_{src}/R_2 = V_b\left(-1/R_5\right) + V_c\left(1/R_2 + 1/R_4 + 1/R_5\right)$$

* Solve the

$$\begin{bmatrix} V_{src}/R_1 \\ V_{src}/R_2 \end{bmatrix} =$$

→ let's Pl

$R_1 = R_2 =$

&

$$\Rightarrow \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} =$$

$$\Rightarrow \begin{bmatrix} 34 \\ 2.8 \end{bmatrix}$$

urrents

R's:

$$\left(\frac{V_2}{R_4}\right)^{0} = 0$$

$$+ 1/R_4)$$

$$)$$

* Solve the linear system:

$$\begin{bmatrix} V_{src}/R_1 \\ V_{src}/R_2 \end{bmatrix} = \begin{bmatrix} 1/R_1 + 1/R_3 + 1/R_5 & -1/R_5 \\ -1/R_5 & 1/R_2 + 1/R_4 + 1/R_5 \end{bmatrix} \begin{bmatrix} V_b \\ V_c \end{bmatrix}$$

→ let's Plug-in Some numbers:

$R_1 = R_2 = R_3 = R_5 = 100\ \Omega$ & $R_4 = 10\ \Omega$

& $V_{src} = 10\ V$

$$\Rightarrow \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.03 & -0.01 \\ -0.01 & 0.12 \end{bmatrix} \begin{bmatrix} V_b \\ V_c \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 34 & 2.8 \\ 2.8 & 8.6 \end{bmatrix} \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} V_b \\ V_c \end{bmatrix} \quad \boxed{\therefore \begin{matrix} V_b = 3.7\ V \\ V_c = 1.1\ V \end{matrix}}$$

Mesh current: Meshes (i.e., unique loops) – A, B, and C

→ Node d set as ground

→ We know $V_a = V_{src}$ and $V_d = 0\,V$

→ Define mesh currents:

$I_A = I_2$, $I_B = I_4$, and $I_C = I_0$

→ Re-write remaining currents in terms of mesh currents:

$I_1 = I_C - I_A$, $I_3 = I_C - I_B$, and $I_5 = I_B - I_A$

* Apply KVL for meshes:

→ A: $V_{R1} - V_{R2} + V_{R5} = 0$

→ B: $-V_{R5} - V_{R4} + V_{R3} = 0$

→ C: $V_{src} - V_{R1} - V_{R3} = 0$

* Use ohm's law to rewrite in terms of mesh currents and R's:

→ A: $V_{R1} -$

$\Rightarrow I_1 R_1$

$\Rightarrow R_1(I$

$\therefore 0 =$

→ B: $-V$

$\Rightarrow -I$

$\Rightarrow R_5$

$\therefore 0 =$

→ C: V_s

$\Rightarrow V_{sr}$

$\Rightarrow V_{sr}$

$\therefore V$

* Solv

$\begin{bmatrix} 0 \\ 0 \\ V_{s} \end{bmatrix}$

Unique loops)

V

I_0

in

$\rightarrow A: V_{R1} - V_{R2} + V_{R5} = 0$

$\Rightarrow I_1 R_1 - I_2 R_2 + I_5 R_5 = 0$

$\Rightarrow R_1(I_C - I_A) - R_2 I_A + R_5(I_B - I_A) = 0$

$\therefore 0 = I_A(R_1 + R_2 + R_5) + I_B(-R_5) + I_C(-R_1)$

$\rightarrow B: -V_{R5} - V_{R4} + V_{R3} = 0$

$\Rightarrow -I_5 R_5 - I_4 R_4 + I_3 R_3 = 0$

$\Rightarrow R_5(I_A - I_B) + R_4(-I_B) + R_3(I_C - I_B) = 0$

$\therefore 0 = I_A(-R_5) + I_B(R_3 + R_4 + R_5) + I_C(-R_3)$

$\rightarrow C: V_{src} - V_{R1} - V_{R3} = 0$

$\Rightarrow V_{src} - I_1 R_1 - I_3 R_3 = 0$

$\Rightarrow V_{src} + R_1(I_A - I_C) + R_3(I_B - I_C) = 0$

$\therefore V_{src} = I_A(-R_1) + I_B(R_3) + I_C(R_1 + R_3)$

* Solve the linear system:

$$\begin{bmatrix} 0 \\ 0 \\ V_{src} \end{bmatrix} = \begin{bmatrix} R_1 + R_2 + R_5 & -R_5 & -R_1 \\ -R_5 & R_3 + R_4 + R_5 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 \end{bmatrix} \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix}$$

let $R_1 = R_2 = R_3 = R_5 = 100\Omega$ & $R_4 = 10\,\Omega$

& $V_{src} = 10\,V$

$$\Rightarrow \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 300 & -100 & -100 \\ -100 & 210 & -100 \\ -100 & -100 & 200 \end{bmatrix} \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0.0091 & 0.0086 & 0.0089 \\ 0.0086 & 0.014 & 0.011 \\ 0.0089 & 0.011 & 0.015 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix}$$

$$\boxed{\therefore I_A = 0.088A,\ I_B = 0.11A,\ \& \ I_C = 0.15A}$$

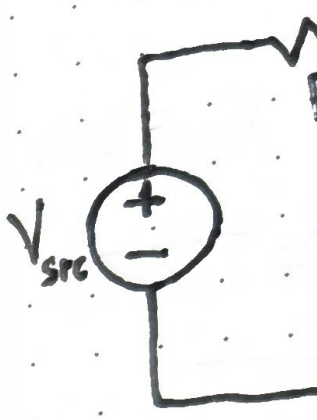
Compare Methods: $I_3 = 0.037A$ & $I_4 = 0.11A$

$V_b = R_3 I_3$ and $V_c = R_4 I_4$

$$\boxed{\therefore V_b = 3.7\,V \text{ and } V_c = 1.1\,V}$$

Same result as Node-Voltage Method ✓

25092



Short-

Using

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Théve

R_{Th} =

∴